

Demographics and Housing Cycles

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How does population aging affect house prices?

- Prior studies have documented that population aging contributes to a long-run downward trend in house prices
 - Bakshi and Chen (1994); Bergantino (1998); Takáts (2010); Gevorgyan (2019)
- However, little is known about whether demographic changes affect the *cyclical fluctuations* in house prices

⇒ **This study examines how and why demographics shape housing price cycles**

Young housing cycle: Young areas tend to experience larger housing cycles

- Using country and US regional panel data, we show that large housing cycles (exceeding 1 st. dev.) are more frequent when areas have younger demographics

Young Expectation: Young people amplify the cycle through long-term overreaction

- Young people show more pronounced 'delayed overreaction' to the market shocks
- Such overreaction amplifies the housing cycle

- **OECD Country Panel Data**

- Number of countries: 29
- Sample period: 1970 - 2021
- Number of observations: 1,263
- Variables: Real housing price, Real GDP per capita, Population by age

- **Freddie Mac US CBSA Panel Data**

- Number of CBSA: 377
- Sample: 1975Q1 - 2020Q1
- Number of observations: 68,237
- Variables: Real housing price, Population by age, Supply elasticity

Bin-scatter plot

- **Bin-scatter plot:** relationship between the *ratio of young population* the *large housing cycle* (two-way fixed effect controlled)

- **Ratio of young population**

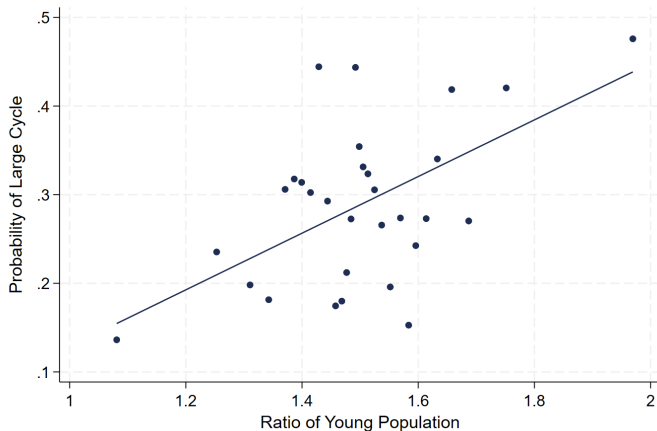
$$= \frac{\text{Population aged 20–39}}{\text{Population aged 50–69}}$$

- **Large housing cycle**

$$= \begin{cases} 1, & \text{if } |\hat{p}_{i,t}| > \sigma_{\hat{p}_i}, \\ 0, & \text{otherwise,} \end{cases}$$

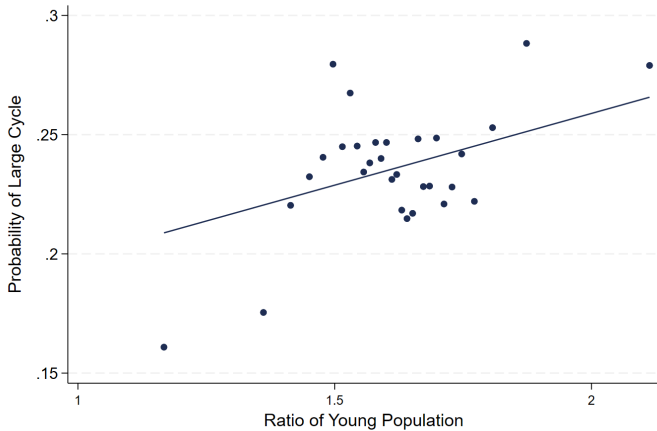
- $\hat{p}_{i,t}$: cyclical component of the log real house price in i at t (HP filtered)
- $\sigma_{\hat{p}_i}$: standard deviation of $\hat{p}_{i,t}$

Bin scatter plot: country panel



Notes: This figure plots the binned relationship between the probability of a large house price cycle and the ratio of the population aged 20–39 to 50–69. The plot is based on 30 equally spaced bins. Both country and time fixed effects are controlled for.

Bin scatter plot: US CBSA panel



Notes: This figure plots the binned relationship between the probability of a large house price cycle and the ratio of the population aged 20–39 to 50–69. The plot is based on 30 equally spaced bins. Both CBSA and time fixed effects are controlled for.

$$\Pr(LHC_{i,t} = 1) = \Lambda(\beta_0 + \beta_1 YO\ ratio_{i,t} + \mathbf{Z}'_{i,t}\boldsymbol{\phi}) ,$$

$$LHC_t = \begin{cases} 1, & \text{if } |\hat{p}_{i,t}| > \sigma_{\hat{p}_i}, \\ 0, & \text{otherwise,} \end{cases}$$

- $\hat{p}_{i,t}$: cyclical component of the log real house price in i at t (HP filtered)
- $\sigma_{\hat{p}_i}$: standard deviation of $\hat{p}_{i,t}$
- $YO\ ratio_{i,t}$: proportion of the population aged 20–39 / 50–69
- $\mathbf{Z}_{i,t}$: vector of controls
- $\Lambda(\cdot)$: logistic cumulative distribution function

Logit model: country panel

	(1)	(2)	(3)	(4)	(5)
Constant	-1.77***	-3.99***	-5.38	-3.11***	-17.28
YO ratio	0.57**	2.11***	2.19***	2.01***	1.88**
(Marginal Effect)	0.11**	0.40***	0.42***	0.36***	0.34**
GDP per Cap.			0.12		1.32
(Marginal Effect)			0.42		0.24
Observation	1,262	1,262	1,225	1,222	1,196
Country FE	No	Yes	Yes	Yes	Yes
Time FE	No	No	No	Yes	Yes
Pseudo R^2	0.0151	0.0630	0.0625	0.1240	0.1217

Notes: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the country level.

Logit model: US CBSA panel

	(1)	(2)	(3)	(4)	(5)
Constant	-1.56***	-1.94***	-2.04***	-1.02***	-0.29*
Supply Elasticity			-0.02**		-0.03**
YO ratio	0.24***	0.47***	0.33***	0.38***	0.13***
(Marginal Effect)	0.04***	0.08***	0.06***	0.05***	0.02***
Observation	67,629	67,629	43,881	66,121	42,417
CBSA FE	No	Yes	No	Yes	No
State FE	No	No	Yes	No	Yes
Time FE	No	No	No	Yes	Yes
Pseudo R^2	0.0035	0.0189	0.0112	0.2640	0.2577

Notes: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the CBSA level.

Amplification of Cycle through Young Expectation

- Delayed Overreaction
 - Literature shows that the price expectation has delayed overshooting
- Young Overreaction
 - We find that young adults (20-39) shows more pronounced delayed overreaction
- Amplification of price cycle through expectation channel
 - We claim that the delayed overreaction stems from Bayesian updating
 - Old agents tend to have a rigid prior on persistent change in price trend
 - We show that a less-rigid prior on persistent price shocks can amplify the price cycle

Expectation Update on Housing price: Literature Review

Adam et al.(2024): Housing price expectation shows 'delayed overreaction'

- Framework

$$X_{t+h+4}/X_{t+h} - \mathbb{E}_{t+h}[X_{t+h+4}/X_{t+h}] = \beta_0 + \beta_1 X_t/X_{t-1} + \epsilon_t^1$$

- $\frac{X_{t+h+4}}{X_{t+h}}$: Annual housing price growth from $t + h$
- $\frac{X_t}{X_{t-1}}$: Quarterly housing price growth from $t - 1$

- Question: Whether agents systematically adjust $\mathbb{E}_{t+h}[X_{t+h+4}/X_{t+h}]$ when there is a price shock

¹Michigan Survey of Consumers(expected HP growth), Case Shiller index (realized HP growth)

Expectation Update on Housing price: Literature Review

Adam et al.(2024): Housing price expectation shows 'delayed overreaction'

$$X_{t+h+4}/X_{t+h} - \mathbb{E}_{t+h}[X_{t+h+4}/X_{t+h}] = \beta_0 + \beta_1 X_t/X_{t-1} + \epsilon_t$$

- interpretation

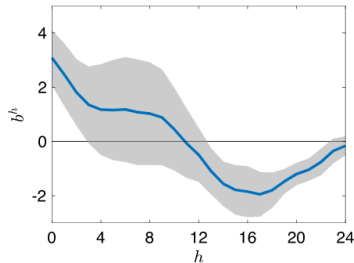
- ① $\beta_1 = 0$: FIRE (Full Information Rational Expectation)
 - Forecast errors should not be predictable (Angelotos et al. (2021))
- ② $\beta_1 > 0$: Underreaction
 - $\mathbb{E}_{t+h}[X_{t+h+4}/X_{t+h}]$ went down when X_t/X_{t-1} goes up
 - Then forecast error becomes systematically positive
- ③ $\beta_1 < 0$: Overreaction
 - vice versa

Expectation Update on Housing price: Literature Review

Adam et al.(2024): Housing price expectation shows 'delayed overreaction'

$$X_{t+h+4}/X_{t+h} - \mathbb{E}_{t+h}[X_{t+h+4}/X_{t+h}] = \beta_0 + \beta_1 X_t/X_{t-1} + \epsilon_t$$

(b) Housing price growth forecast errors



Result

- ① In the short term, updates are **sluggish** to the observed change in housing prices
- ② Around 3 years, agents become **overly optimistic (pessimistic)** about housing price growth when the price-to-rent ratio is **high (low)**

Heterogeneous Expectation Updates by Age Group

We extend the former regression model with **demographics interaction term**

$$X_{t+h+12}/X_{t+h} - \mathbb{E}_{t+h}^i [X_{t+h+12}/X_{t+h}] = \beta_0 + \beta_1 Dummy_{2039} \\ + \beta_2 X_t/X_{t-1} + \beta_3 Dummy_{2039} \cdot X_t/X_{t-1} + \epsilon_t^2$$

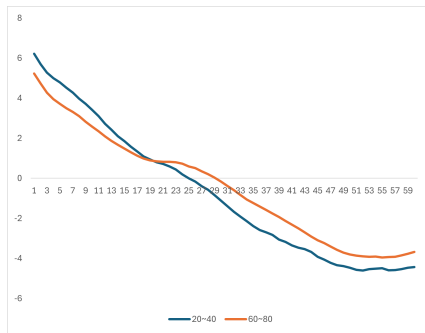
- Hypothesis: Overreaction is more pronounced among the Young

Or, Equivalently $\beta_3 < 0$, when $\beta_2 < 0$

²Monthly data, Michigan Survey of Consumers(expected HP growth), Case Shiller index (realized HP growth), No. of obs : 47,797, Time : 2007.1-2024.12

Heterogeneous Expectation Updates by Age Group

$$X_{t+h+12}/X_{t+h} - \mathbb{E}_{t+h}^i [X_{t+h+12}/X_{t+h}] = \beta_0 + \beta_1 Dummy_{2039} \\ + \beta_2 X_t/X_{t-1} + \beta_3 Dummy_{2039} \cdot X_t/X_{t-1} + \epsilon_t$$



- **Result:** $\beta_3 < 0$ for $h > 19$

Greater overreaction in long-term expectation among young agents

Model: Bayesian Updates

We present a **Bayesian updating model** to explain the source of overreaction

- Decomposition of shocks: $\frac{X_t}{X_{t-1}} = b_t + \epsilon_t$
 - Transitory shocks: $\epsilon_t \sim N(0, \sigma_\epsilon)$
 - Persistent shocks: $b_{t+1} = b_t + \nu_t$, $\nu_t \sim N(0, \sigma_\nu)$
- Source of delayed overreaction
 - In short run, interpret the price shock as temporary
 - In the long term, it gets interpreted as persistent shocks
- Demographics and Bayesian updates
 - Aged agents have more informed-prior: rigid expectation on persistent component
 - The story is coherent to inflation expectation literature (e.g. Malmendier and Nagel (2015))

Simulation: Expectation updates and Housing cycle

We construct a model to connect the prior difference and the housing cycle

- Model Setting

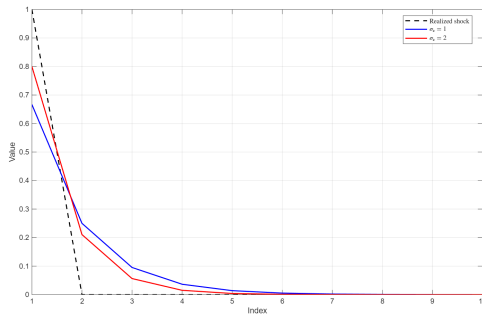
- ① Agents observe realized housing price growth $\{y_\tau\}_{\tau=0}^T$
- ② State space form: $y_t = b_t + \epsilon_t$, $b_t = b_{t-1} + \nu_t$
- ③ $\epsilon_t \sim N(0, \sigma_\epsilon)$, $\nu_t \sim N(0, \sigma_\nu)$
- ④ While assuming $\sigma_\epsilon = 1$, we simulate the model for different σ_ν
- ⑤ Agents use **Kalman Filter** to update the expectation

We compare agents with different σ_ν to see how the expectation update changes

Primary Simulation Result

- Setting

- 1 Shock structure: one-time shock at $t = 1$
- 2 We compare Young ($\sigma_\nu = 1$) and Old ($\sigma_\nu = 2$)



- Interpretation

- $\sigma_\nu = 1$ reacts sluggishly in $t = 1$ but overreacts afterwards
- Belief on stable persistent components (low σ_ν) produces delayed over response, which was observed in the empirical results